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TREND-BASED MOCK PAPER

GATE Life Sciences

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Life Sciences
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Life Sciences

Q1. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q2. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q3. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q4. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q5. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q6. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q7. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q8. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q9. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q10. [1-NAT] [1 mark]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q13. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q16. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q18. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(\frac{4}{5}, \frac{2}{5}, 0)$
 - (B) $(\frac{8}{5}, \frac{4}{5}, 0)$
 - (C) $(2, 1, 0)$
 - (D) $(1, 2, 3)$
-

Q19. [2-MSQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
 - (B) 2
 - (C) 4
 - (D) 1
-

Q20. [2-MSQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q21. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q22. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q23. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q24. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q25. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q26. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q27. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q28. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q29. [1-MCQ] [1 mark]

Grignard reacts with:

- (A) Alcohol only
- (B) CO₂, carbonyl compounds
- (C) Acids only
- (D) Alkanes

Q30. [1-MCQ] [1 mark]

Geometrical isomerism in:

- (A) 1-butene
- (B) 2-butene
- (C) 1-butyne
- (D) 2-butyne

Q31. [1-NAT] [1 mark]

Hofmann elimination gives:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Benzene undergoes:

- (A) Only addition
- (B) Only substitution
- (C) Both
- (D) Elimination

Q33. [1-NAT] [1 mark]

SN₂ rate depends on:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MCQ] [2 marks]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q35. [2-NAT] [2 marks]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-NAT] [2 marks]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MSQ] [2 marks]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q38. [2-MCQ] [2 marks]

Grignard reacts with:

- (A) Alcohol only
- (B) CO₂, carbonyl compounds
- (C) Acids only
- (D) Alkanes

Q39. [2-MSQ] [2 marks]

Geometrical isomerism in:

- (A) 1-butene
- (B) 2-butene
- (C) 1-butyne
- (D) 2-butyne

Q40. [2-MCQ] [2 marks]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q41. [1-MCQ] [1 mark]

Strongest field ligand (spectrochemical series):

- (A) I⁻
- (B) Br⁻
- (C) CN⁻
- (D) F⁻

Q42. [1-MCQ] [1 mark]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q43. [1-MCQ] [1 mark]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q44. [1-NAT] [1 mark]

[Co(NH₃)₆]³⁺ coordination number:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MCQ] [2 marks]

Octahedral crystal field splits d-orbitals into:

- (A) t_{2g} and e_g
- (B) e_g and t_{2g} only
- (C) s and p

(D) sigma and pi

Q46. [2-NAT] [2 marks]

[Co(NH₃)₆]³⁺ coordination number:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Octahedral crystal field splits d-orbitals into:

- (A) t_{2g} and e_g
- (B) e_g and t_{2g} only
- (C) s and p
- (D) sigma and pi

Q48. [2-MCQ] [2 marks]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q49. [2-MCQ] [2 marks]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q50. [2-MCQ] [2 marks]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q51. [2-MCQ] [2 marks]

Octahedral crystal field splits d-orbitals into:

- (A) t_{2g} and e_g
- (B) e_g and t_{2g} only
- (C) s and p
- (D) sigma and pi

Q52. [1-MCQ] [1 mark]

dG = -5kJ/mol at 300K. Equilibrium constant:

- (A) e²
- (B) e⁻²
- (C) 2
- (D) 0.5

Q53. [1-MCQ] [1 mark]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

- (A) $\sim 110\text{ kJ/mol}$
- (B) $\sim 56\text{ kJ/mol}$
- (C) $\sim 22\text{ kJ/mol}$
- (D) $\sim 8\text{ kJ/mol}$

Q54. [1-NAT] [1 mark]

pH of 0.01M HCl:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

Van der Waals corrects for:

- (A) Only volume
- (B) Only pressure
- (C) Both volume and pressure
- (D) Temperature

Q56. [1-MCQ] [1 mark]

First-order: 50% in 10 min. 90% time:

- (A) 10 min
- (B) 33.2 min
- (C) 20 min
- (D) 5 min

Q57. [2-NAT] [2 marks]

pH of 0.01M HCl:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-NAT] [2 marks]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

(Numerical Answer Type — Enter the numerical value)

Q59. [2-NAT] [2 marks]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

(Numerical Answer Type — Enter the numerical value)

Q60. [1-NAT] [1 mark]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q62. [1-NAT] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q64. [1-MCQ] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q65. [2-MCQ] [2 marks]

Strongest acid:

- (A) HCl
 - (B) HNO_3
 - (C) H_2SO_4
 - (D) HClO_4
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	D	1
2	1MCQ	A	1
3	1MCQ	D	1
4	1MCQ	C	1
5	1MCQ	B	1
6	1NAT	A	1
7	1MCQ	A	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1NAT	B	1
12	1MCQ	B	1
13	1NAT	B	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1NAT	B	1
17	1NAT	C	1
18	1MCQ	B	1
19	1MCQ	A	1
20	1MCQ	B	1
21	1MCQ	C	1
22	1MCQ	C	1
23	1MCQ	B	1
24	1MCQ	B	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1MCQ	C	1
28	1MCQ	A	1

29	1NAT	A	1
30	1NAT	A	1
31	2NAT	A	2
32	2NAT	C	2
33	2NAT	A	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2NAT	B	2
37	2MCQ	A	2
38	2MCQ	B	2
39	2MSQ	C	2
40	2MSQ	B	2
41	2MCQ	B	2
42	2MCQ	C	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	A	2
46	2NAT	C	2
47	2MCQ	B	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2NAT	B	2
51	2MCQ	A	2
52	2MCQ	B	2
53	2NAT	B	2
54	2MSQ	B	2
55	2MCQ	B	2
56	2NAT	B	2
57	2MSQ	B	2
58	2MCQ	A	2

59	2MCQ	A	2
60	2NAT	B	2
61	2MCQ	A	2
62	2NAT	B	2
63	2MCQ	A	2
64	2MCQ	B	2
65	2MCQ	D	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q2.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x \cdot v$ and $y(1) = 0$ gives $v = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q3.

Answer: D

Solution:

Very dilute: [H⁺] from water dominates. [H⁺]_{total} approx 1.1e-7. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q4.

Answer: C

Solution:

CN⁻ (strongest) > NH₃ > en > NO₂⁻ > F⁻ > Cl⁻ > Br⁻ > I⁻ (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q5.

Answer: B

Solution:

Grignard: reacts with CO₂ (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q6.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q7.

Answer: A

Solution:

$dG = -RT \ln(K) \Rightarrow K = \exp(-dG/RT) = \exp(5000/(8.314 \cdot 300)) = e^2.$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q8.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q9.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310)). E_a = R \ln(2)/(1/300 - 1/310) = 53.6 \text{ kJ/mol. Closest: 56.}$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q10.

Answer: A

Solution:

Low-spin d⁶: t_{2g}⁶, all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q11.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: difficult

Q12.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q13.

Answer: B

Solution:

SN2: bimolecular. Rate = $k[\text{substrate}][\text{nucleophile}]$. Both affect rate.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q14.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q15.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q16.

Answer: B

Solution:

$[\text{H}^+] = 10^{-2}$. $\text{pH} = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q17.

Answer: C

Solution:

6 NH_3 ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q18.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1 + 4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: B

Solution:

$\Delta_n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q21.

Answer: C

Solution:

$(P + a/V^2)(V - b) = RT$. Corrects for intermolecular forces (a) and finite volume (b).

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q22.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: B

Solution:

$t_{90} = t_{50} * \log(10)/\log(2) = 10 * 3.32 = 33.2$ min.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q24.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}*[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Octahedral: t_2g (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: $d_{x^2-y^2}, d_{z^2}$).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q36.

Answer: B

Solution:

$f(0)=2, f(1)=0, f(-1)=4, f(2)=4, f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: B

Solution:

$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q40.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: B

Solution:

$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1, B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: C

Solution:

6 NH₃ ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q47.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: d_{x²-y²}, d_{z²}).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q49.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: difficult

Q50.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q51.

Answer: A

Solution:

Low-spin d₆: t_{2g}⁶, all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q52.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q53.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q54.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q55.

Answer: B

Solution:

Grignard: reacts with CO₂ (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q56.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q57.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q58.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310))$. $E_a = R \ln(2)/(1/300 - 1/310) = 53.6$ kJ/mol. Closest: 56.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q61.

Answer: A

Solution:

Low-spin d⁶: t_{2g}⁶, all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q62.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310))$. $E_a = R \ln(2)/(1/300 - 1/310) = 53.6$ kJ/mol. Closest: 56.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q63.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: d_{x²-y²}, d_{z²}).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q64.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q65.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 28 questions, 45 marks
Organic Chemistry: 12 questions, 19 marks
Inorganic Chemistry: 11 questions, 18 marks
Physical Chemistry: 8 questions, 11 marks
Chemistry: 6 questions, 7 marks

Difficulty Distribution:

Easy: 21 | Moderate: 36 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.